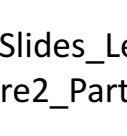


Slides_Lecture2_PartitionsMethods_M2_AdvDataSciences_2026

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Advanced Data Sciences

Partition-based methods

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Lecture 2, M2 - Maths/Info & PSND
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Empirical risk minimization (ERM)

Notation recall : ERM

- Classifier/Prediction function : $g : \mathcal{X} \rightarrow \mathcal{Y}$
- Loss function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbf{R}_+$
- Training sample $\mathcal{D}_n = \{(X_i, Y_i), i = 1, \dots, n\}$
- Risk and empirical risk of g
$$R_P(g) = \mathbb{E}_P[\ell(Y, g(X))], \quad \hat{R}_n(g) = \frac{1}{n} \sum_{i=1}^n \ell(Y_i, g(X_i))$$
- Main classifiers
 - Bayes classifier
$$g_P^* = \arg \min_g R_P(g)$$
 - Optimal classifier in $\mathcal{G} \subset \mathcal{F}(\mathcal{X}, \mathcal{Y})$
$$g_{\mathcal{G}, P}^* = \arg \min_{g \in \mathcal{G}} R_P(g)$$
 - Empirical Risk Minimizer (ERM)
$$\hat{g}_{n, \mathcal{G}} \in \arg \min_{g \in \mathcal{G}} \hat{R}_n(g), \quad \text{with } \mathcal{G} \subset \mathcal{F}(\mathcal{X}, \mathcal{Y})$$

General VC-bound for binary classification

ERM for partition-based methods

ERM for partition-based methods - general result

Particular case 1 : k-Nearest Neighbors

Particular case 1 : k-Nearest Neighbors - partition

Particular case 1 : k-Nearest Neighbors - Voronoi regions

Particular case 1 : k-Nearest Neighbors - a consistency result

Particular case 2 : Trees

Particular case 2 : Trees - technical details

Particular case 2 : Trees - expansion

Particular case 2 : Trees - illustration for CART

A Side Note on Random Forests^a

Bagging

Bagging

Key arguments in Bagging

$g^*(x) = 1$
 $\eta(x) \geq \frac{1}{2}$
where $\eta(x) = P(Y=1|X=x)$

